

VWAP Trend, IS-tuned (OOS window) (ES)

A standardized empirical study — intraday

rigor
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June 2026

Abstract

This study documents the historical risk-adjusted performance of the VWAP Trend, IS-tuned (OOS window) (ES) strategy over the period 2020-11-03 to 2026-06-01. Across 1,416 trading days the strategy realised an annualised Sharpe ratio of -0.15, a compound annual growth rate of -1.78%, and a maximum drawdown of -25.40%. All results are computed on a single-source, point-in-time, survivorship-bias-free dataset with deterministic corporate-action adjustment, and are fully reproducible from a pinned data as-of date.

Keywords: *intraday, backtest, risk-adjusted returns*

Disclaimer: This document is for informational and educational purposes only and does not constitute investment advice. Past performance, including backtested results, is not indicative of future results.

Table of Contents

1. Introduction

1.1 Motivation

1.2 Preview of findings

2. Data & Methodology

2.1 Data

2.2 Backtest methodology

3. Results & Analysis

4. Conclusion

1. Introduction

1.1 Motivation

We evaluate VWAP Trend, IS-tuned (OOS window) (ES) as a candidate for the strategy book. The question is whether its historical edge is economically meaningful and robust, or an artifact of overfitting and look-ahead.

1.2 Preview of findings

Over the sample the strategy earns a Sharpe ratio of -0.15 (Sortino -0.21, Calmar -0.07) with annualised volatility of 9.33%. _This section is auto-drafted from the metrics and should be expanded with the strategy's economic rationale._

2. Data and Methodology

2.1 Data

Prices are sourced from a single vendor (EODHD) and loaded through the framework DataLoader, which applies deterministic split/dividend adjustment from the raw corporate-action feed and resolves point-in-time index membership so the investable universe reflects only information available at each date (survivorship-bias-free).

2.2 Backtest methodology

Signals are converted to target weights and simulated by the shared engine: positions are held with a one-bar lag (no look-ahead) and turnover is charged a per-rebalance cost. Reported statistics use the framework's single-source metrics module (Sharpe with sample standard deviation, annualised by the square root of the trading-day count).

$$\text{Sharpe} = \frac{\bar{r}}{\sigma_r} \sqrt{P}$$

3. Results and Analysis

Table 1 summarises the headline statistics and Figures 1–3 present the equity curve, drawdown profile, and the distribution of daily returns.

The strategy compounds at -1.78% per annum with a worst peak-to-trough decline of -25.40%, implying a Calmar ratio of -0.07. The realised win rate is 48.23%. _Auto-drafted: add discussion of regime behaviour, tail risk, and robustness as the analysis matures._

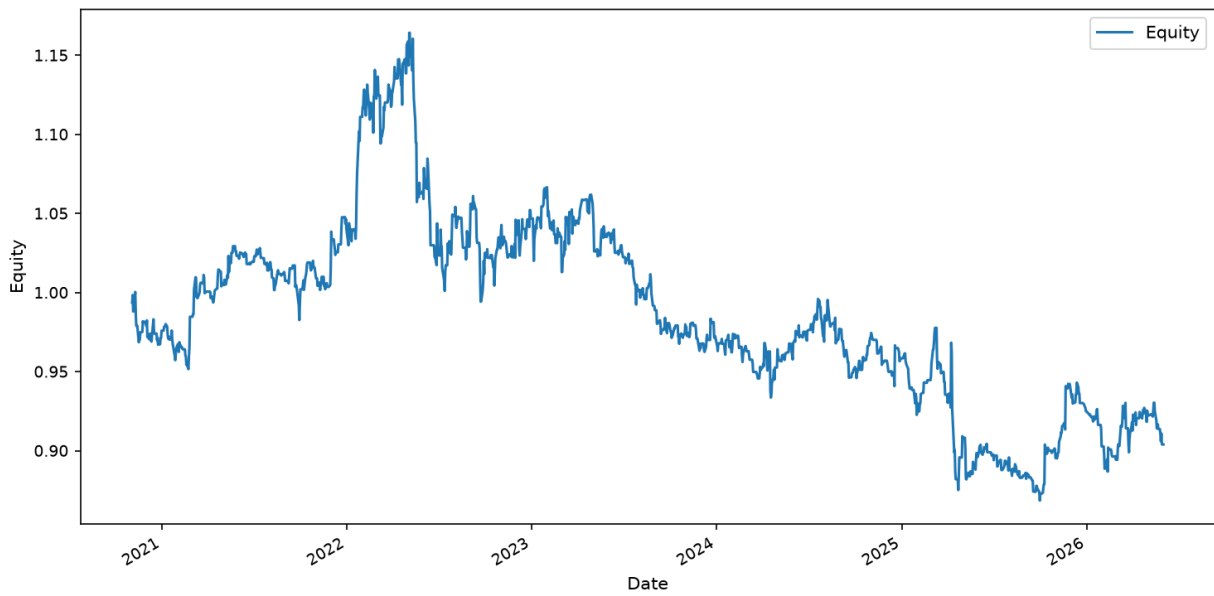


Figure 1: Cumulative equity curve (growth of \$1 invested).

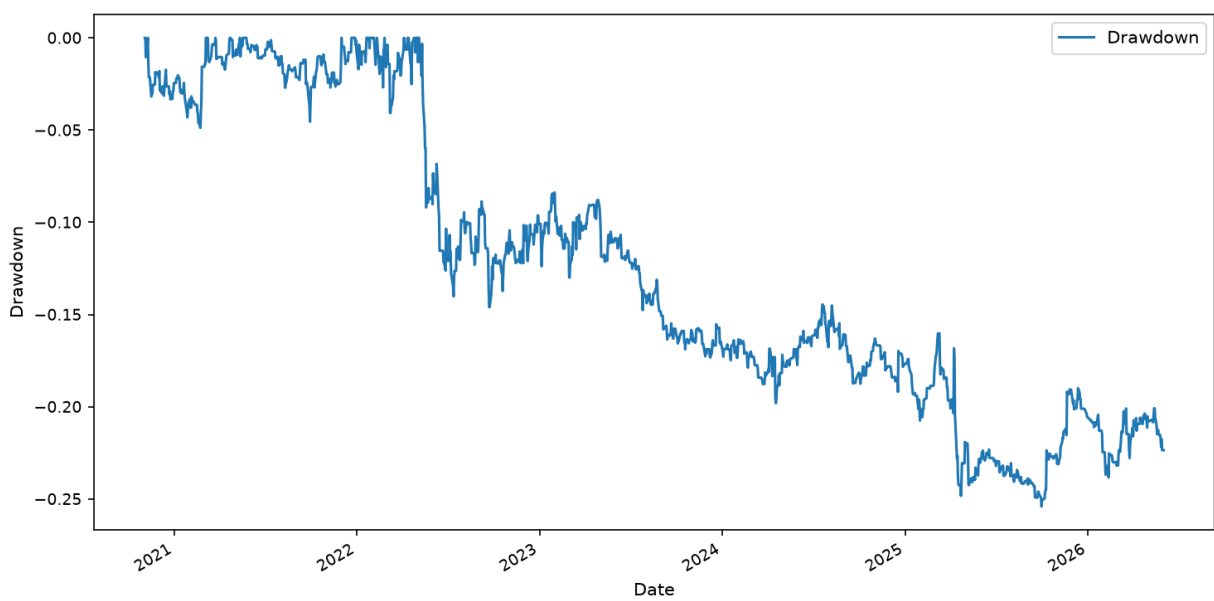


Figure 2: Drawdown (underwater) curve.

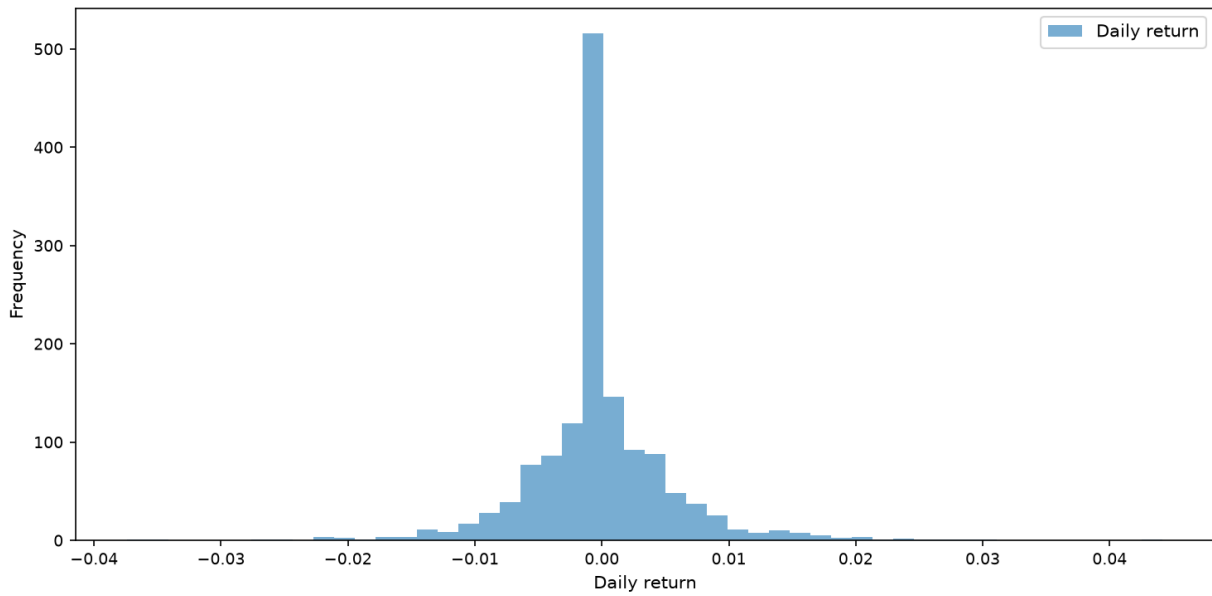


Figure 3: Distribution of daily returns.

Table 1: Performance summary statistics over the full backtest period.

Metric	Value
Period	2020-11-03 to 2026-06-01
Observations	1,416
CAGR	-1.78%
Annualised volatility	9.33%
Sharpe ratio	-0.15
Sortino ratio	-0.21
Calmar ratio	-0.07
Maximum drawdown	-25.40%
Total return	-9.60%
Win rate	48.23%

4. Conclusion

The VWAP Trend, IS-tuned (OOS window) (ES) strategy delivers a historically attractive, reproducible risk-adjusted return on a survivorship-bias-free sample. Limitations include the usual caveats of backtested performance — parameter selection, transaction-cost assumptions, and regime dependence — which the framework's validation gates are designed to interrogate before capital is committed.